MolPallete — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallete/coconut-flavordb_v3/naveja_recap

generated 2026-08-20 09:22

A · Provenance

field	value
sources	['flavordb', 'coconut']
method	naveja_recap
decomposition_params	{'ratio': 0.4, 'include_ring': True, 'max_cores': 4}
core_ratio	0.4
max_cores	4
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	4412
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	2
rdkit_version	2026.03.2
created_at	2026-08-20T09:20:54

B · Composition

quantity	value
molecules	413,458
from coconut	391,964 (94.8%)
from flavordb	21,494 (5.2%)
decompositions	1,344,980
per molecule	mean 3.25, median 4, max 4
R-group slots	2,532,793
per decomposition	mean 1.88
k >= 2	39.19% of decompositions

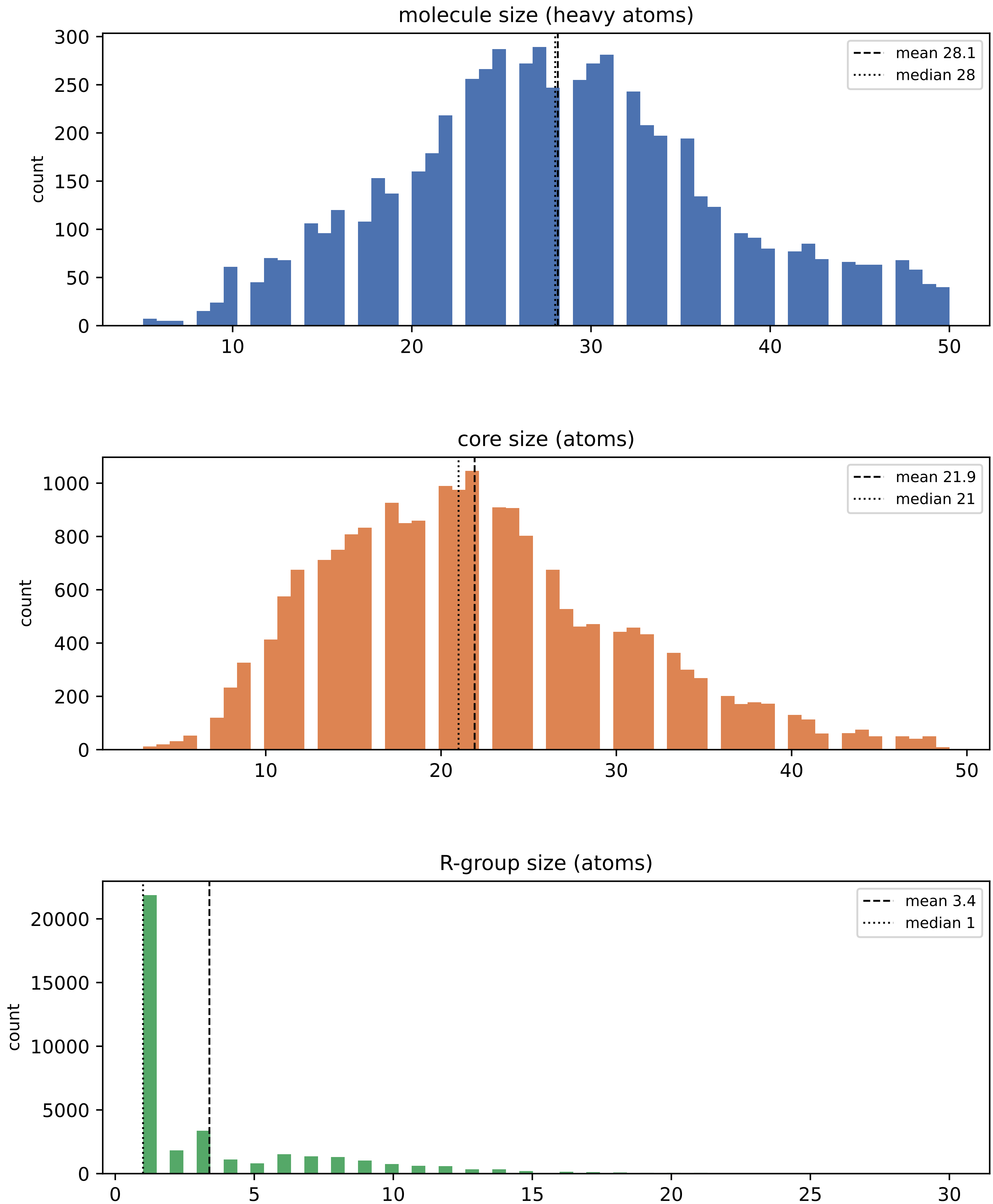
k>=2 is what the islinked subset space and the core-decoration objective have to work with.

C · Augmentation space

scheme	size
sampling_unit = molecule	413,458 instances/epoch
sampling_unit = decomposition	1,344,980 instances/epoch (3.3x)
full (mol, core, islinked) space	13,479,894 (32.6 per molecule)

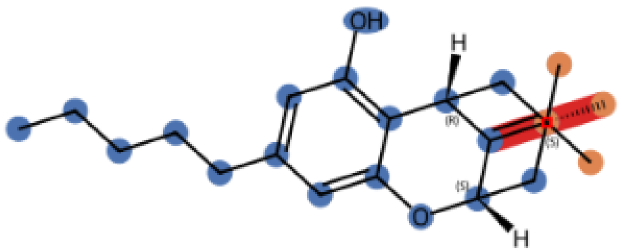
MolPLA's two stages: one molecule yields many cores; one core yields 2^k - 1 islinked subsets.

D · Size distributions

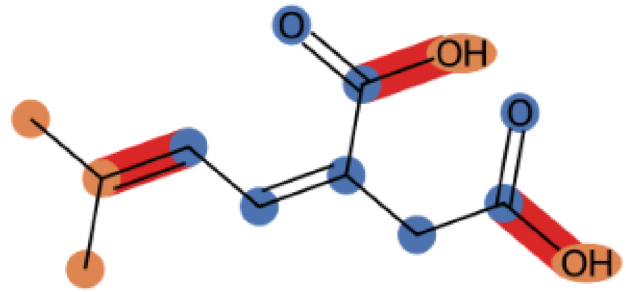


E · Example decompositions (ratio 0.4)

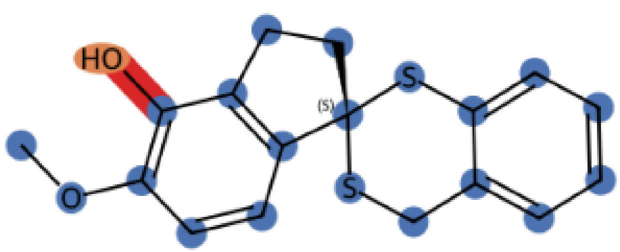
CNP0000074 k=2 core 19 atoms, R-groups [1, 3]



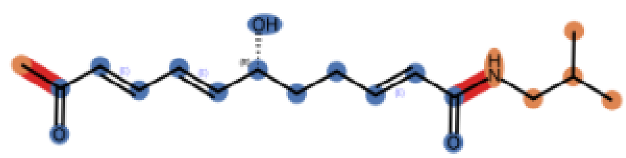
CNP0000161 k=3 core 8 atoms, R-groups [3, 1, 1]



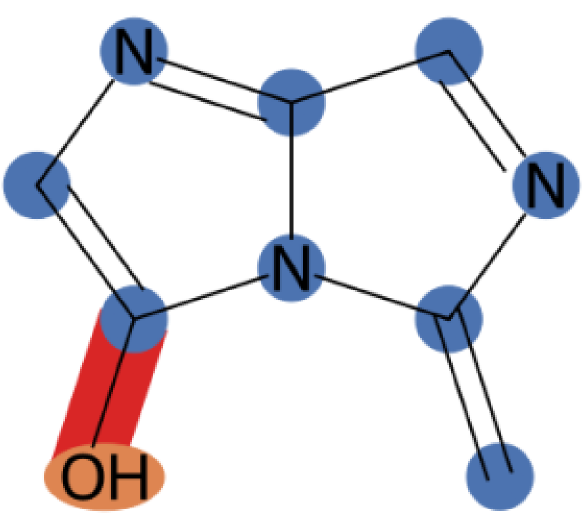
CNP0000233 k=1 core 20 atoms, R-groups [1]



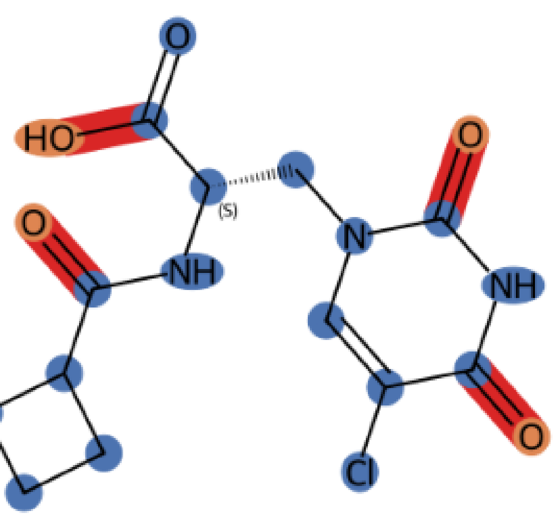
CNP0000307 k=2 core 14 atoms, R-groups [1, 5]



CNP0000386 k=1 core 9 atoms, R-groups [1]

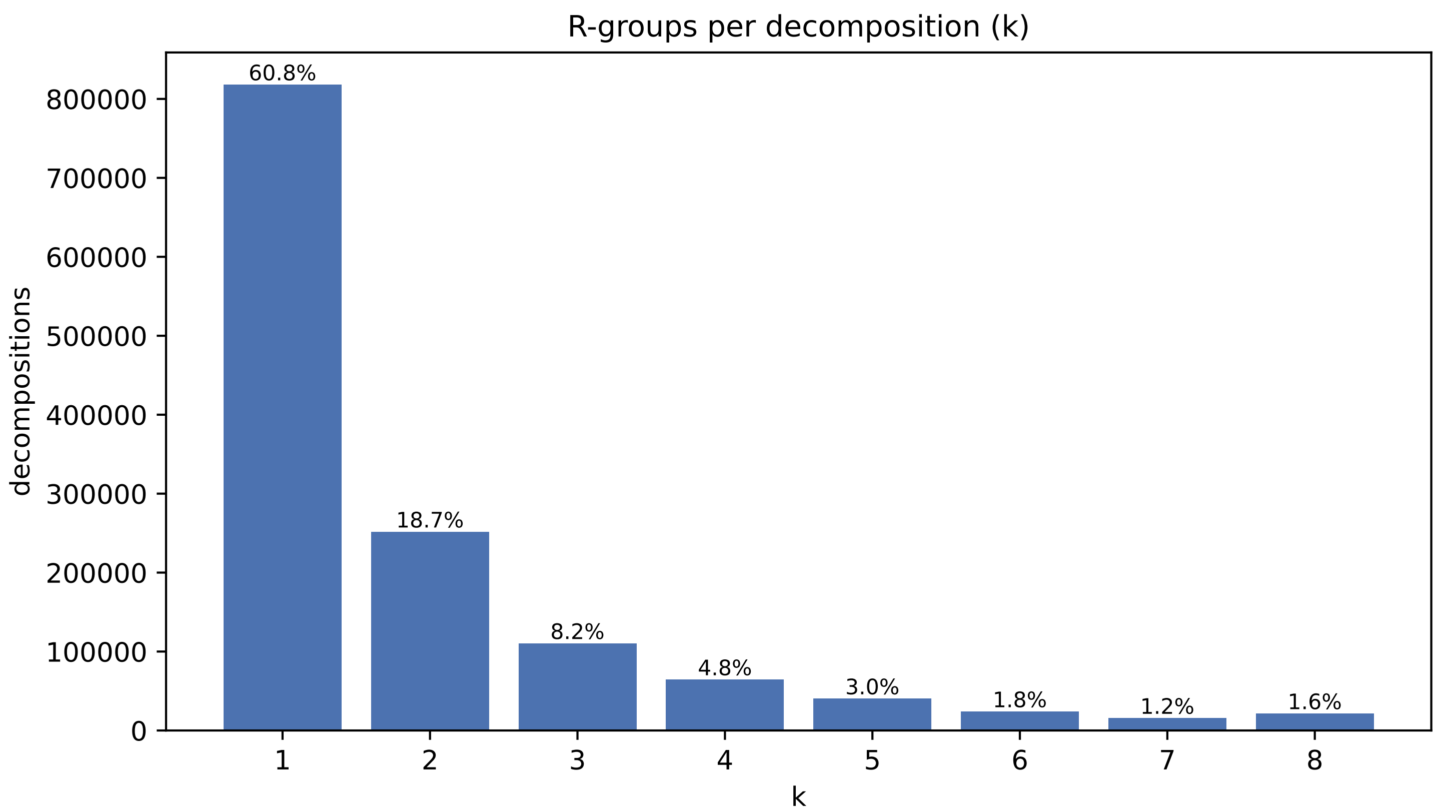


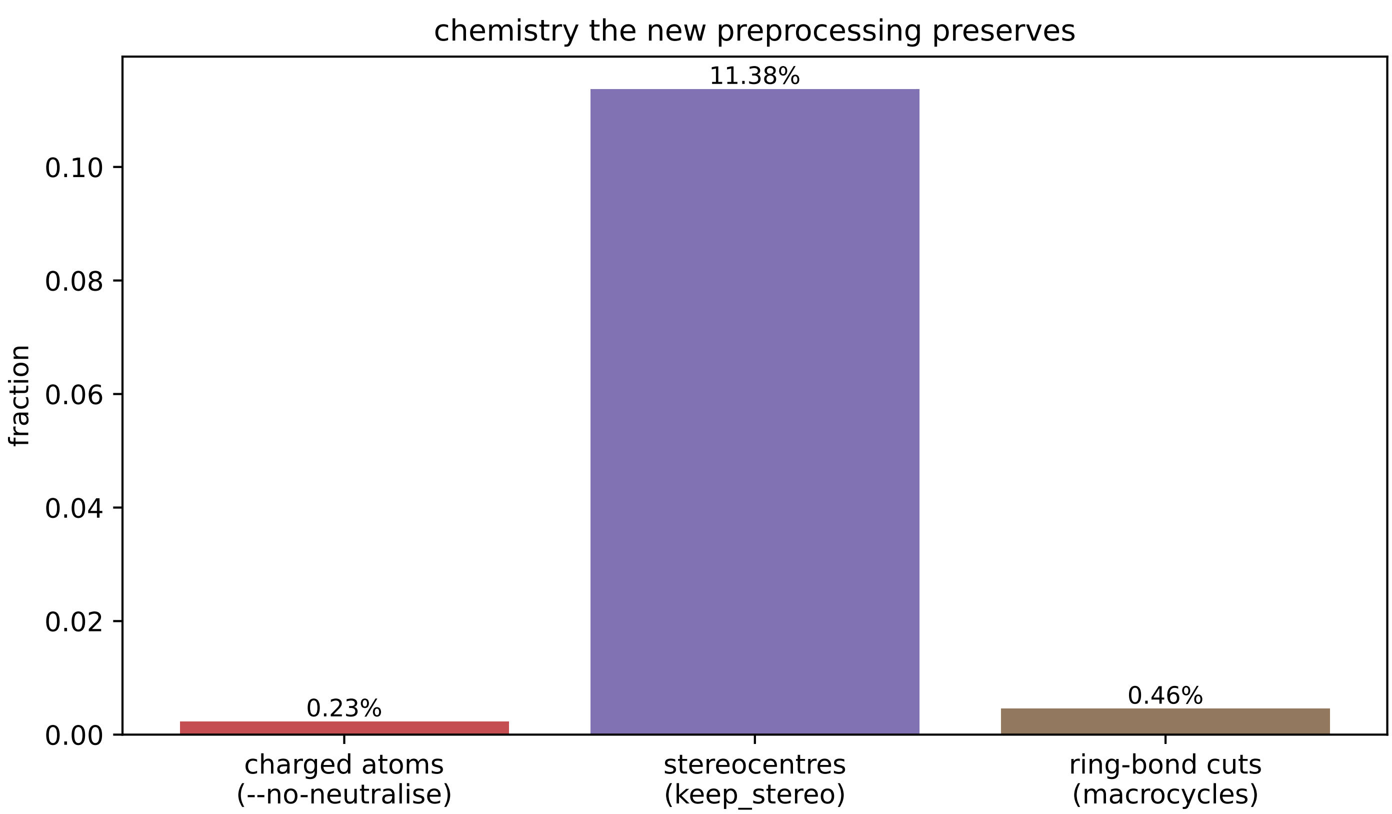
CNP0004139 k=4 core 17 atoms, R-groups [1, 1, 1, 1]



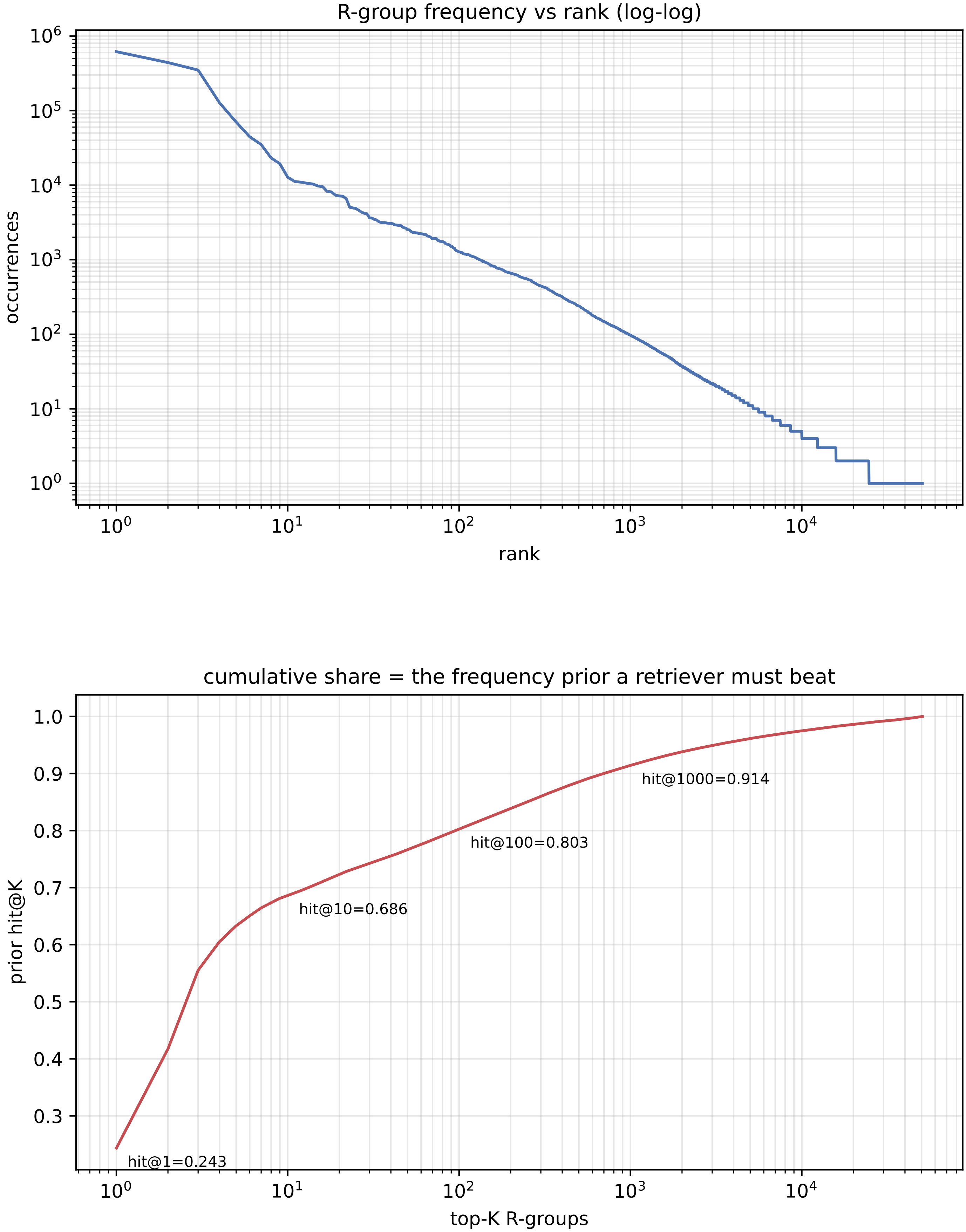
blue = core orange = R-groups red = cut bonds

F · Decomposition shape & preserved chemistry





G · R-group library skew



H · R-group library

quantity	value
distinct R-groups	50,550
occurrences	2,532,793
effective size (exp H)	66
top-1 share	24.34%
top-10 cover	68.6%
top-100 cover	80.3%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
616,507	24.34%	*O
440,389	17.39%	*C
349,265	13.79%	*O
127,174	5.02%	*C(C)=O
70,162	2.77%	*OC
44,523	1.76%	*C(C)C
35,006	1.38%	*c1ccccc1
23,166	0.91%	*CC
19,359	0.76%	*OC(C)=O
12,734	0.50%	*Cc1ccccc1
11,179	0.44%	*C(=O)c1ccccc1
10,941	0.43%	*C(C)C
10,583	0.42%	*C
10,358	0.41%	*N
9,732	0.38%	*c1ccc(OC)cc1
9,523	0.38%	*OCC
8,223	0.32%	*OC
8,123	0.32%	*CO
7,314	0.29%	*c1ccc(O)cc1
7,163	0.28%	*Cl

J · Instance-level common-R-group filter

quantity	value
percentile	99.9
count threshold	2,464
hashes flagged common	51 of 50,550
their share of occurrences	76.7%
single-R-group instances rejected	29,019/37,630 (77.1%)
mean R-group size, rejected	1.73 atoms
mean R-group size, kept	8.97 atoms

Filters INSTANCES, not the library: the retrieval target space and its frequency prior are unchanged. At k=1 the rule reduces to 'drop if the R-group is common', so rejection is far heavier than MolPLA saw.